1 Methods

1.1 Synthetic orientation maps

We generated 40,000 two-dimensional polycrystalline microstructures with DREAM.3D [1], each a 128×128 pixel orientation map on a square grid with a step of 1 μm and a single cubic phase (nickel, point group $m\bar{3}m$) [grain-size distribution and texture settings used in DREAM.3D]. A map contains on average 50 grains (standard deviation 4), the mean equivalent circle diameter of a grain is about 20 pixels, and 17% of the pixels lie on a grain boundary, which we define as a pixel with at least one of its four neighbours in a different grain. Two maps were incomplete because their generation run was interrupted and were discarded, leaving 39,998. Since every map is a separate microstructure, we split the data by map: maps 1–500 form the test set, maps 501–1000 the validation set, and the remaining 38,998 maps the training set. No grain therefore appears in more than one set. All maps were produced with the same generator settings, and the number of grains per map is the same across the three ranges.

1.2 Noise model

Each noisy map combines two kinds of error. The first is orientation scatter: every pixel is rotated by a small rotation about an axis drawn uniformly on the sphere, with a rotation angle drawn uniformly between 0 and $\theta_{\max}$. The disorientation between a scattered pixel and its true orientation therefore has a mean of $\theta_{\max}/2$. The second is misindexing: a fraction p of the pixels is replaced by orientations drawn uniformly from SO(3) [2], so that these pixels carry no information about their true orientation. Because misindexing in real scans occurs mostly where the diffraction pattern mixes two grains, a boundary pixel is chosen four times as often as an interior pixel, sampled without replacement. Scatter is applied first and misindexing second. All errors are measured as the disorientation angle, the smallest rotation between two orientations once the 24 proper rotations of the cubic group are taken into account.

The network is trained on noise that is drawn afresh every time a training map is used. For each draw, $\theta_{\max}$ is sampled uniformly from $[0°, 15°]$ and p from $[0, 10\%]$, so the network never sees the same corruption twice and is never told the noise level of its input. We also apply a random rotation of the specimen frame to all orientations of the map, which leaves every misorientation between pixels unchanged, together with random flips and 90° rotations of the pixel grid. The 500 validation maps were corrupted once, with a level drawn per map from the same ranges and a fixed random seed, so that every epoch and every training run is compared on identical inputs. The validation set is used only to select the checkpoint.

1.3 Test conditions

The test maps were corrupted under 28 fixed conditions: every combination of $\theta_{\max} \in \{1, 3, 5, 7, 9, 12, 15\}°$ and $p \in \{0, 3, 6, 9\}\%$, each applied to all 500 test maps (Fig. 1). The lowest level gives a mean error of 0.5°, comparable to the angular precision usually reported for Hough-based indexing [3]; at $\theta_{\max} = 15°$ (mean 7.5°) the grains are only just distinguishable. All 28 conditions were generated with the same random seed. For a given map, the rotation axes and the fraction of $\theta_{\max}$ drawn for each pixel are thus the same in every condition, and the scatter is identical across the four misindexing levels, so any difference between two conditions comes from the amount of noise and not from a different random draw. At $\theta_{\max} \geq 12°$, scatter alone moves 17–33% of the pixels beyond 10° from their true orientation. We therefore assess the repair of misindexed pixels on the recorded set of replaced pixels rather than with a fixed angular threshold.

2 Results

2.1 Measuring the denoising error

To compare a denoised map with the clean map we need the angle between two orientations. For a cubic crystal the same orientation can be written in 24 equivalent ways, so a plain difference between two sets of Euler angles is not a valid measure [4, 5]. We therefore use the disorientation angle. Let $\mathbf{q}$ and $\mathbf{q}'$ be the unit quaternions of two orientations. Their misorientation is

$$\Delta\mathbf{q} = \mathbf{q}' \, \mathbf{q}^{-1}, \tag{1}$$

and the disorientation angle is the smallest rotation angle among the symmetrically equivalent forms of $\Delta\mathbf{q}$,

$$\delta(\mathbf{q}, \mathbf{q}') = \min_{\mathbf{s} \in \mathcal{S}} \; 2 \arccos \left| (\mathbf{s} \, \Delta\mathbf{q})_0 \right|, \tag{2}$$

where $\mathcal{S}$ is the set of the 24 proper rotations of the cubic point group and $(\cdot)_0$ is the scalar part of a quaternion.

For a map m with the set of pixels Ω ($|\Omega| = 128 \times 128 = 16{,}384$), denoised map u_m and true map u_m^0, the denoising error is the mean per-pixel disorientation,

$$e_m = \frac{1}{|\Omega|} \sum_{\mathbf{x} \in \Omega} \delta\big(u_m(\mathbf{x}), u_m^0(\mathbf{x})\big). \tag{3}$$

We also compute the same mean over two subsets of the pixels: the grain-boundary pixels B_m, which have at least one of their four neighbours in a different grain of the true map, and the misindexed pixels M_m, which the noise generator replaced. We call the three measures ALL (e_m), BOUNDARY (e_m^B) and CORRUPTED (e_m^M). Last, the percentage of pixels that are still grossly wrong is

$$g_m = \frac{100}{|\Omega|} \sum_{\mathbf{x} \in \Omega} \mathbb{1}\Big[\delta\big(u_m(\mathbf{x}), u_m^0(\mathbf{x})\big) > 10°\Big], \tag{4}$$

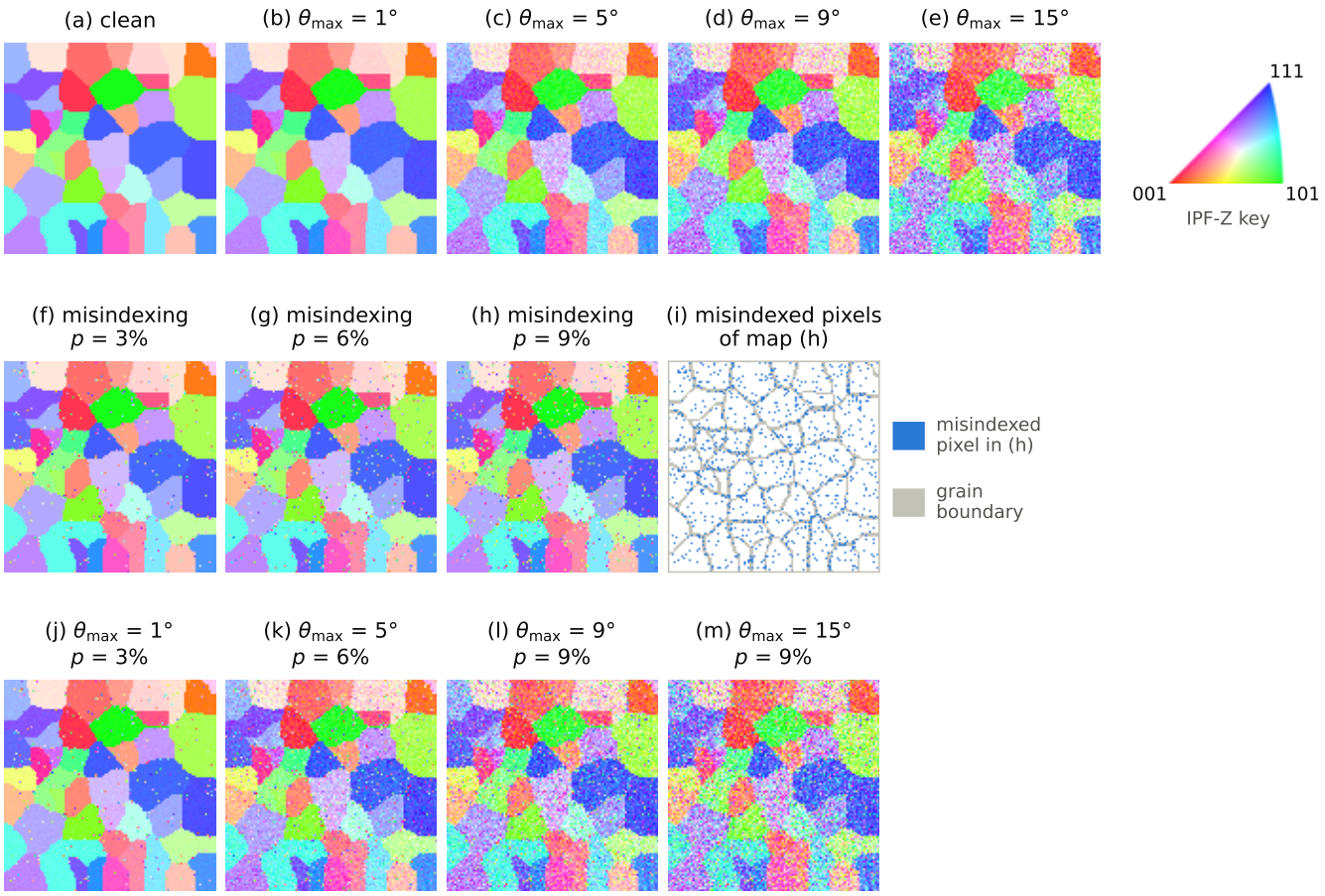


Figure 1: The noise model on test map 1, coloured with the IPF-Z key. (a) Clean map. (b)–(e) Orientation scatter alone, with maximum angle $\theta_{\max} = 1°$, 5°, 9° and 15°. (f)–(h) Misindexing alone, $p = 3$, 6 and 9% of the pixels: the replaced pixels of the corresponding test sets shown on the clean map, without their scatter. (i) Positions of the misindexed pixels in (h) (blue) relative to the grain boundaries (grey): boundary pixels make up 16% of the map but receive 41% of the misindexing, because a boundary pixel is four times as likely to be chosen. (j)–(m) Scatter and misindexing together, from the mildest to the most severe test condition: $\theta_{\max} = 1°$ with $p = 3\%$, 5° with 6%, 9° with 9%, and 15° with 9%.

where $\mathbb{1}[\cdot]$ is 1 when the condition holds and 0 otherwise.

Each test set has $N = 500$ maps. For each set we report the mean over the maps, $\bar{e} = \frac{1}{N}\sum_{m=1}^{N} e_m$. Since every map has the same number of pixels, $\bar{e}$ is also the mean disorientation over all $N|\Omega|$ pixels of the set. We trained the network three times, with different random seeds. Unless stated otherwise, the value given for our method is the mean of the three runs, and the ± value is the standard deviation across the runs, which shows how much a result depends on the random start of training. To compare noise levels we also use the fraction of the input error that the denoising removed,

$$R = 1 - \frac{\bar{e}_{\text{out}}}{\bar{e}_{\text{in}}}. \tag{5}$$

To test whether two methods differ, we use the paired t-test on the per-map errors [6], as Atindama et al. [4] did. For the same n maps let $d_m = e_m^{(1)} - e_m^{(2)}$ be the difference between the errors of the two methods on map m, $\bar{d}$ the mean and S_d the standard deviation of the d_m. The test statistic

$$T = \frac{\bar{d}}{S_d/\sqrt{n}} \tag{6}$$

has a t-distribution with $n - 1$ degrees of freedom. We take $p < 0.05$ as statistically significant.

2.2 Training

Figure 2 shows the training and validation loss of the three runs. Each run had 60 epochs of 2,437 batches of 16 maps, about 5.8 min per epoch and 5.8 h per run on one NVIDIA H100 GPU. The training and validation losses fall together and stay close at every epoch, so the network does not overfit the training maps. Both flatten after about epoch 45. For each run we kept the checkpoint

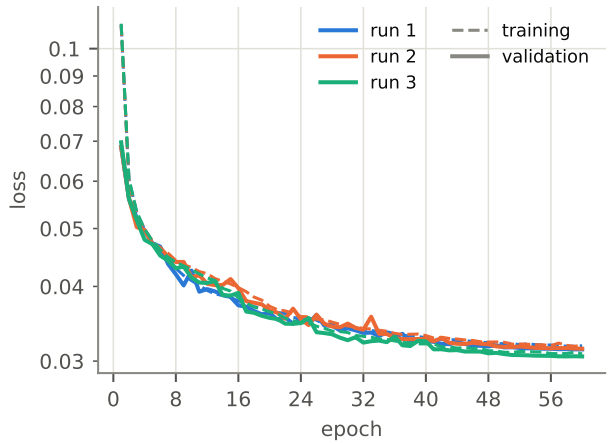


Figure 2: Training loss (dashed) and validation loss (solid) of the three training runs. The two losses stay close throughout, which shows that the network does not overfit the training maps.

Table 1: Mean per-pixel errors on the mixed test set (500 maps, each with its own random noise level). For our method, mean ± standard deviation over three training runs. R is the fraction of the error removed, Eq. (5).

Measure	Noisy input	Ours	R
ALL (°)	5.692	0.589 ± 0.015	89.6%
BOUNDARY (°)	8.554	1.197 ± 0.020	86.0%
CORRUPTED (°)	40.709	2.466 ± 0.032	93.9%
$> 10°$ (% pixels)	11.27	0.226 ± 0.002	98.0%

with the best validation score, which came from epochs 58, 50 and 58. On the validation maps the gate, the part of the network that marks a pixel as misindexed, found 98.8% of the misindexed pixels (recall), and 99.9% of the pixels it marked were truly misindexed (precision). The three runs gave the same values to the third decimal.

2.3 Maps with mixed noise levels

We first report one overall result. In the mixed test set, each of the 500 test maps has its own noise level, drawn at random from the same ranges as in training: $\theta_{\max}$ between 0° and 15° and p between 0 and 10%. Table 1 shows the four error measures. The mean per-pixel error drops from 5.69° to $0.589 \pm 0.015°$, so 89.6% of the error is removed. The misindexed pixels, which are 40.7° from their true orientation on average, end 2.47° from it, and the percentage of pixels more than 10° wrong falls from 11.27% to 0.23%.

2.4 Effect of the noise level

Tables 2 and 3 give the results for the 28 test conditions, and Figure 3 plots the errors against the maximum scatter angle. The standard deviation across the three runs is at most 0.032° for the mean error and 0.006 percentage points for the percentage of gross errors, far smaller than any of the differences discussed below.

The error left after denoising grows in proportion to the scatter. With scatter alone ($p = 0$), the mean error goes from 0.068° at $\theta_{\max} = 1°$ to 0.922° at $\theta_{\max} = 15°$, but the fraction removed stays between 86.5% and 88.1% at every level. This is what we expect from averaging. The network replaces each pixel by a weighted average of its neighbours in the same grain, and averaging random errors shrinks them by a fixed factor, whatever their size. So the remaining error is a fixed fraction of the input error. When misindexing is added, the fraction removed is larger, between 88.1% and 93.4%, because the misindexed pixels carry a large part of the input error and most of them are repaired.

With misindexing, the percentage of grossly wrong pixels (Table 3) falls from about p in the input to between 0.10% and 0.58% after denoising. At $\theta_{\max} = 12°$ and 15°, scatter alone puts 16.7% and 33.4% of the pixels beyond 10°; after denoising only 0.02% and 0.05% remain.

2.5 Repair of misindexed pixels

A misindexed pixel carries no information about its true orientation, so it must be rebuilt from its neighbours. Over the 21 test conditions with misindexing, between 91.7% ($\theta_{\max} = 15°$, $p = 9\%$) and 95.5% ($\theta_{\max} = 1°$, $p = 3\%$) of the misindexed pixels end within 5° of their true orientation. Their mean error drops from about 40.7° to between 2.02° and 3.26° (Fig. 3, right). The repair becomes slightly harder as the scatter grows. A likely reason is that a misindexed pixel whose random orientation happens to lie close to the true one is hard to tell apart from a strongly scattered pixel.

2.6 Where the remaining errors are

Figure 4 shows one test map under three noise conditions. The denoised maps are close to the true map of Fig. 1(a), even at $\theta_{\max} = 15°$ where the grains in the input are hard to see. In the error maps of the denoised maps, the remaining error follows the grain boundaries. The numbers confirm this. In every test condition, between 92.5% and 99.7% of the pixels that remain more than 10° wrong lie on a grain boundary, although boundary pixels make up only 17% of a map. With misindexing, most of these pixels (72–98%) are misindexed pixels on a boundary. Such a pixel has neighbours from two grains, and the network sometimes gives it the orientation of the wrong grain. For the same reason the BOUNDARY error is always higher than the ALL error (Fig. 3, middle).

Table 2: Mean per-pixel error, ALL (°), for the 28 test conditions, 500 maps each: maximum scatter angle $\theta_{\max}$ and percentage of misindexed pixels p. Ours is the mean of three training runs (standard deviation across runs $\leq 0.032°$).

	$p = 0\%$		$p = 3\%$		$p = 6\%$		$p = 9\%$	
$\theta_{\max}$ (°)	Input	Ours	Input	Ours	Input	Ours	Input	Ours
1	0.500	0.068	1.708	0.128	2.913	0.196	4.121	0.273
3	1.500	0.186	2.679	0.248	3.853	0.317	5.031	0.396
5	2.501	0.301	3.649	0.365	4.793	0.437	5.942	0.518
7	3.501	0.418	4.619	0.485	5.733	0.560	6.852	0.645
9	4.501	0.538	5.589	0.609	6.673	0.688	7.762	0.777
12	6.002	0.724	7.044	0.803	8.084	0.891	9.127	0.989
15	7.502	0.922	8.500	1.013	9.494	1.111	10.492	1.222

Table 3: Percentage of pixels more than 10° from their true orientation, Eq. (4), for the same conditions as Table 2. Ours is the mean of three training runs (standard deviation across runs ≤ 0.006 percentage points).

	$p = 0\%$		$p = 3\%$		$p = 6\%$		$p = 9\%$	
$\theta_{\max}$ (°)	Input	Ours	Input	Ours	Input	Ours	Input	Ours
1	0.00	0.00	2.98	0.10	5.96	0.23	8.94	0.37
3	0.00	0.00	2.98	0.11	5.96	0.23	8.94	0.38
5	0.00	0.00	2.98	0.11	5.96	0.24	8.94	0.39
7	0.00	0.00	2.98	0.12	5.96	0.25	8.94	0.41
9	0.00	0.00	2.98	0.13	5.96	0.27	8.94	0.43
12	16.66	0.02	19.14	0.15	21.62	0.31	24.10	0.49
15	33.35	0.05	35.33	0.21	37.30	0.38	39.29	0.58

2.7 Comparison with MTEX filters (preliminary)

So far the MTEX filters [7] have been run on only the first 10 test maps of two scatter-only conditions, $\theta_{\max} = 5°$ and 7°, with their default parameters. Table 4 compares them with our method on these maps. Our method has a lower mean error than the mean, median, Kuwahara, half-quadratic and infimal-convolution filters on all 10 maps at both levels, and the paired t-tests give $p < 0.001$. The spline filter is the closest: its mean error is 0.403° against our 0.300° at $\theta_{\max} = 5°$, and 0.601° against 0.416° at $\theta_{\max} = 7°$. However, the spline error changes a lot from map to map (standard deviation 0.23° and 0.30°, against 0.009° and 0.012° for ours). Our method is lower on 6 and 7 of the 10 maps, and with only 10 maps the difference is not statistically significant ($p = 0.17$ and $p = 0.08$). A full comparison on all 500 test maps, with filter parameters tuned for each noise level and with misindexed pixels, is in progress.

Table 4: Mean per-pixel error ALL (°) on test maps 1–10, scatter only: mean (standard deviation across the 10 maps). Ours is averaged over three training runs. p-values of the paired t-test against our method: *** $p < 0.001$; n.s. not significant ($p \geq 0.05$).

Method	$\theta_{\max} = 5°$	$\theta_{\max} = 7°$
Noisy input	2.502	3.503
Mean	0.986 (0.016) ***	1.383 (0.017) ***
Median	0.408 (0.014) ***	0.575 (0.019) ***
Kuwahara	0.977 (0.013) ***	1.371 (0.017) ***
Spline	0.403 (0.226) n.s.	0.601 (0.301) n.s.
Half-quadratic	1.824 (0.022) ***	2.569 (0.030) ***
Infimal convolution	1.916 (0.015) ***	2.903 (0.022) ***
Ours	**0.300** (0.009)	**0.416** (0.012)

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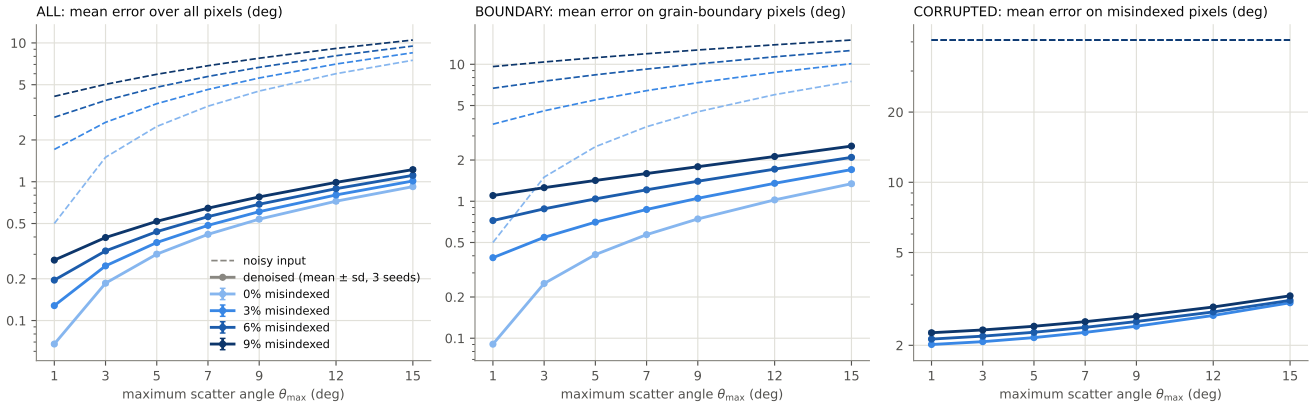


Figure 3: Mean per-pixel error against the maximum scatter angle, for each percentage of misindexed pixels: (left) all pixels, (middle) grain-boundary pixels, (right) misindexed pixels. Dashed lines: noisy input. Solid lines with markers: our method, mean ± standard deviation over three training runs (the error bars are smaller than the markers). Logarithmic vertical axes.

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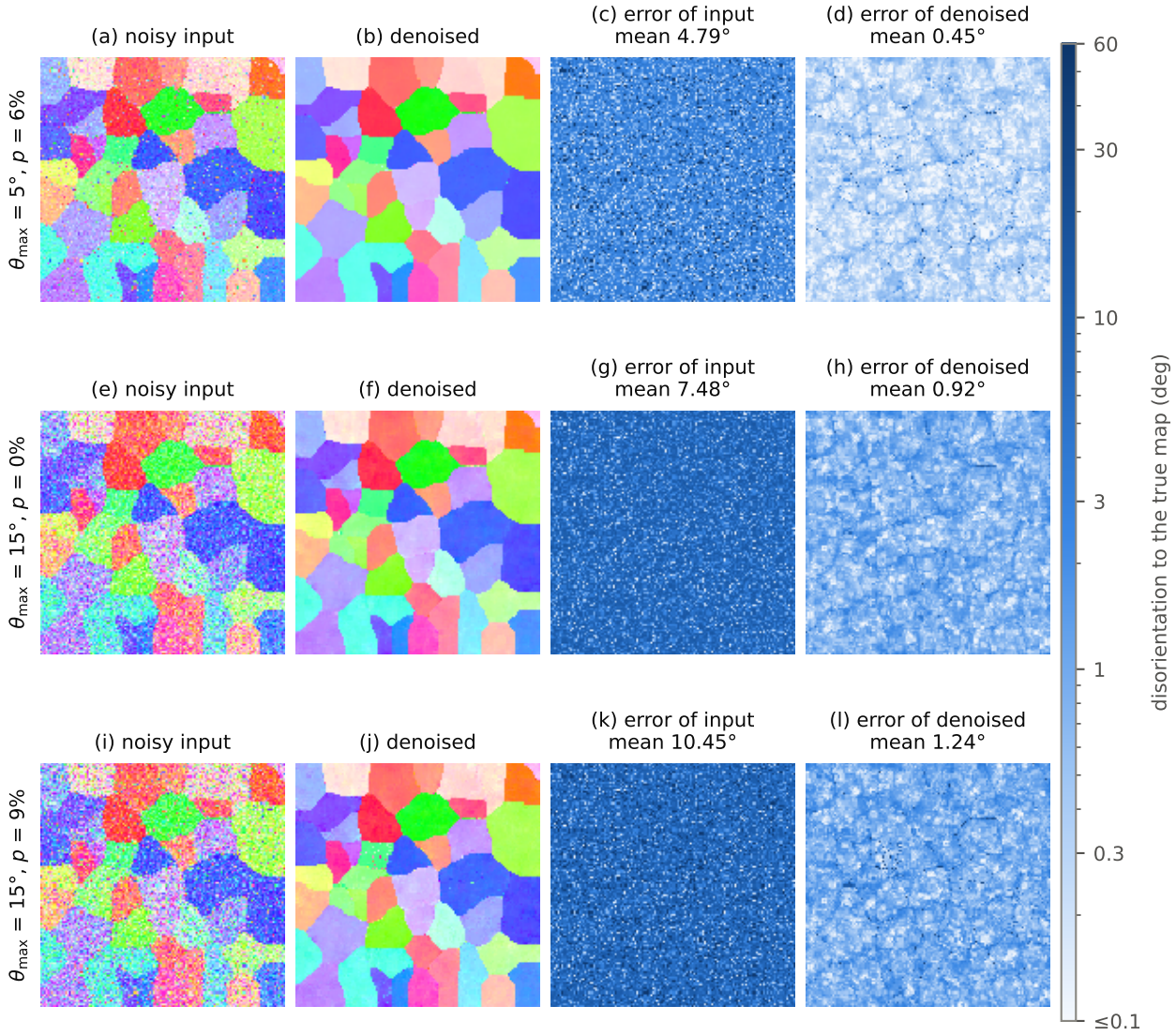


Figure 4: Test map 1 (Fig. 1(a)) denoised by our method (run 1) under three noise conditions, one per row. From left: noisy input and denoised map (IPF-Z colours), and the disorientation of each to the true map (logarithmic colour scale). The mean per-pixel error of each map is given above it. The remaining error follows the grain boundaries.